07 Combine tables with bind_cols() and bind_rows()

Join tables side by side, or stack them one under another.

BY THE END OF THIS SECTION YOU WILL BE ABLE TO

- Add columns from another table with bind_cols()
- Stack rows from another table with bind_rows()
- Spot NA where columns do not match

The R code in this lesson

```
Main_Question <- Dataset %>% select('3. Did you receive 50 KG UREA?',
                                    '4. If not, how many KGs did you receive?',
                                    '10. How was the physical quality of the wheat
seed received?')

# Combining data frames: column-wise (same number of rows)
demographic <- bind_cols(demographic, Main_Question)

# Combining data frames: row-wise (stacks rows)
demographic <- bind_rows(demographic, Initial_Information)
```

What you will hear (times count from the end of the intro)

Intro Data often comes in pieces. In this lesson, you will learn to put the pieces together: side by side with bind columns, or one under another with bind rows. You will also see what NA means when the pieces don't match perfectly. Let's build a bigger picture.

0:00 We have three tables now. Demographic, with five columns. Initial Information, also with five columns. And Main Question, with three survey questions: urea, kilograms received, and seed quality.

0:20 Let's look at demographic first. One hundred thirty five rows, five columns.

0:36 I write a note: combining data frames, column-wise. That means side by side.

1:00 Demographic, arrow, bind underscore cols, and inside, demographic and Main Question. The tooltip helps us with the arguments.

1:12 Run it. Demographic now has eight variables. Open it, and you see the three question columns added on the right. Same rows, more columns.

1:40 Bind columns only works well when both tables have the same rows, in the same order. Here, both come from the same dataset, so they match.

1:52 Now row-wise. Demographic, arrow, bind underscore rows, with demographic and Initial Information.

2:05 Run it. Now we have two hundred seventy rows. The second table was stacked underneath.

2:25 Scroll down, and look at the question columns. In the lower rows, they say NA. NA means not available. Initial Information did not have those columns, so R leaves them empty.

2:55 Bind columns adds columns, bind rows adds rows. And when you see NA, ask yourself: is something missing, or did the tables just not match? Very well done!